

Scalability & Future Plan

Date	7 July 2026
Team ID	xxxxxxx
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

The platform is built to scale with growing student demand and evolving academic needs. In the coming years, it will incorporate advanced educational AI models, real-time analytics, and cloud infrastructure to enhance learning performance, user experience, and support for a larger student audience.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited to local JSON history logs	User progress cannot sync across devices or browsers.	High
2	Dependency on third-party Gemini APIs	API downtime or rate limits may affect learning experience.	Medium
3	Basic text/markdown explanation	Harder to understand for purely visual or auditory learners.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent local users.	Deploy to cloud server (AWS/GCP) with auto-scaling.
2	Data Storage	SQLite (local storage).	Migrate to PostgreSQL or cloud database (Supabase).
3	Performance	Synchronous endpoint handling.	Integrate Redis caching and async background processing.
4	Security	Basic authentication & config keys.	Implement OAuth2, JWT, and encrypted credentials.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Mobile Application (Android & iOS)	6 Months	Improve accessibility and student engagement.
Phase 2	Voice-enabled Chat Tutor (Text-to-Speech)	12 Months	Enhance learning accessibility for diverse learners.
Phase 3	Institutional Curriculums Sync	18 Months	Align quizzes and topics directly with schools/colleges.
Phase 4	Multilingual & Global Accent Support	24 Months	Reach a broader global student audience.